SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%

Code of Conduct:

I T Bhupal / 192425243 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: bhupalreddy

Criteria	**Weightage (%)**	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Understanding of Scenario	20%				
Identification of Key Issues	20%				
Application of Concepts	25%				
Decision Making	20%				
Clarity of Response	15%				

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → **/24**

Quality Assurance → **/25**

Human Resources → **/26**

Executive Management → **/27**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

3.A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the **/64** prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.
2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.
4. The **default route** that should be used if the destination does not match any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

QUESTION 1: Subnet Allocation for a Smart Campus

Given Data

Network Block: 192.168.100.0/24

School of Computing: /25

School of Electronics: /26

School of Mechanical Engineering: /27

Step 1: Calculate Usable Host Addresses

Formula:

Total IP addresses = 2^(32 − Prefix Length)

Usable hosts = Total IP addresses − 2

School of Computing (/25)

Host bits = 32 − 25 = 7

Total IP addresses = 2^7 = 128

Usable hosts = 128 − 2 = 126

School of Electronics (/26)

Host bits = 32 − 26 = 6

Total IP addresses = 2^6 = 64

Usable hosts = 64 − 2 = 62

School of Mechanical Engineering (/27)

Host bits = 32 − 27 = 5

Total IP addresses = 2^5 = 32

Usable hosts = 32 − 2 = 30

Step 2: Determine Network Details

School of Computing (/25)

- Network Address: 192.168.100.0
- Valid Host IP Range: 192.168.100.1 – 192.168.100.126
- Broadcast Address: 192.168.100.127
- Usable Hosts: 126

School of Electronics (/26)

- Network Address: 192.168.100.128
- Valid Host IP Range: 192.168.100.129 – 192.168.100.190
- Broadcast Address: 192.168.100.191
- Usable Hosts: 62

School of Mechanical Engineering (/27)

- Network Address: 192.168.100.192
- Valid Host IP Range: 192.168.100.193 – 192.168.100.222
- Broadcast Address: 192.168.100.223
- Usable Hosts: 30

Step 3: Remaining Unused IP Address Range

The allocated network block is 192.168.100.0/24.

The last assigned subnet ends at 192.168.100.223.

Remaining unused IP address range: **192.168.100.224 – 192.168.100.255**

Final Answer

Department	Network Address	Valid Host Range	Broadcast Address	Usable Hosts
Computing (/25)	192.168.100.0	192.168.100.1–126	192.168.100.127	126
Electronics (/26)	192.168.100.128	192.168.100.129–190	192.168.100.191	62
Mechanical (/27)	192.168.100.192	192.168.100.193–222	192.168.100.223	30

Conclusion: The three departments have 218 usable host addresses in total. The remaining unused range is 192.168.100.224–192.168.100.255.

QUESTION 2: Subnet Allocation for a Multinational Software Company

Given Data

Network Block: 10.10.0.0/16

Software Development: /24

Quality Assurance: /25

Human Resources: /26

Executive Management: /27

Subnets are allocated sequentially using VLSM.

Step 1: Calculate Usable Host Addresses

Software Development (/24)

Host bits = 32 − 24 = 8

Total IP addresses = 2^8 = 256

Usable hosts = 256 − 2 = 254

Quality Assurance (/25)

Host bits = 32 − 25 = 7

Total IP addresses = 2^7 = 128

Usable hosts = 128 − 2 = 126

Human Resources (/26)

Host bits = 32 − 26 = 6

Total IP addresses = 2^6 = 64

Usable hosts = 64 − 2 = 62

Executive Management (/27)

Host bits = 32 − 27 = 5

Total IP addresses = 2^5 = 32

Usable hosts = 32 − 2 = 30

Step 2: Determine Network Details

Software Development (/24)

- Network Address: 10.10.0.0
- Valid Host IP Range: 10.10.0.1 – 10.10.0.254
- Broadcast Address: 10.10.0.255
- Usable Hosts: 254

Quality Assurance (/25)

- Network Address: 10.10.1.0
- Valid Host IP Range: 10.10.1.1 – 10.10.1.126
- Broadcast Address: 10.10.1.127
- Usable Hosts: 126

Human Resources (/26)

- Network Address: 10.10.1.128
- Valid Host IP Range: 10.10.1.129 – 10.10.1.190
- Broadcast Address: 10.10.1.191
- Usable Hosts: 62

Executive Management (/27)

- Network Address: 10.10.1.192
- Valid Host IP Range: 10.10.1.193 – 10.10.1.222
- Broadcast Address: 10.10.1.223

- Usable Hosts: 30

Step 3: Remaining Unused IP Address Range

The original network block extends from 10.10.0.0 to 10.10.255.255.

The last assigned subnet ends at 10.10.1.223.

Remaining unused IP address range: **10.10.1.224 – 10.10.255.255**

Final Answer

Department	**Network Address**	**Valid Host Range**	**Broadcast Address**	**Usable Hosts**
Software Development (/24)	10.10.0.0	10.10.0.1–254	10.10.0.255	254
Quality Assurance (/25)	10.10.1.0	10.10.1.1–126	10.10.1.127	126
Human Resources (/26)	10.10.1.128	10.10.1.129–190	10.10.1.191	62
Executive Management (/27)	10.10.1.192	10.10.1.193–222	10.10.1.223	30

Conclusion: The four departments use 512 IP addresses, including network and broadcast addresses. The remaining unused range is 10.10.1.224–10.10.255.255.

QUESTION 3: IPv6 Address Compression, Expansion and Network Prefix

Given Data

IPv6 Address: 2001:0db8:abcd:0000:0000:0000:0000:0000/64

Prefix Length: /64

Step 1: Compress the IPv6 Address

Rules:

1. Leading zeros in each hexadecimal group can be removed.
2. Consecutive zero groups can be replaced with ::, used only once in an address.

Original Address:

2001:0db8:abcd:0000:0000:0000:0000:0000

After removing leading zeros:

2001:db8:abcd:0:0:0:0:0

After compression:

2001:db8:abcd::/64

Step 2: Expand the Compressed Address

Compressed Address: 2001:db8:abcd::/64

An IPv6 address contains eight hexadecimal groups. The double colon represents five zero groups.

Expanded Address:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

Step 3: Identify the Network Prefix

Total IPv6 address length = 128 bits

Network prefix length = 64 bits

Interface identifier length = 128 − 64 = 64 bits

Network Prefix: **2001:db8:abcd:0::/64**

Step 4: Calculate the Total Number of Possible Host Addresses

Total possible interface identifiers = 2^(128 − Prefix Length)

= 2^(128 − 64)

= 2^64

= **18,446,744,073,709,551,616**

IPv6 does not use broadcast addresses. This is the theoretical number of possible 64-bit interface identifier values.

Final Answer

- Compressed Address: 2001:db8:abcd::/64
- Expanded Address: 2001:0db8:abcd:0000:0000:0000:0000:0000/64
- Network Prefix: 2001:db8:abcd:0::/64
- Interface Identifier Length: 64 bits
- Total Possible Interface Identifier Values: 18,446,744,073,709,551,616

Conclusion: The given IPv6 address compresses to 2001:db8:abcd::/64. The first 64 bits identify the network, and the remaining 64 bits provide the interface identifier space.

QUESTION 4: Routing Table Analysis and Packet Forwarding

Given Data

Routing Table Entries:

1. 192.168.1.0/24
2. 192.168.2.0/24
3. 192.168.3.0/24

Destination IP Address: 192.168.2.45

Step 1: Determine the Destination Subnet

Destination IP Address = 192.168.2.45

Subnet mask for /24 = 255.255.255.0

The first three octets identify the network.

Therefore, the destination subnet is **192.168.2.0/24**.

Step 2: Identify the Matching Routing Table Entry

The destination IP address 192.168.2.45 matches the entry:

192.168.2.0/24

The router selects this route to forward the packet.

Step 3: Determine the Next-Hop Network or Interface

The matching route is 192.168.2.0/24.

The router forwards the packet through the outgoing interface or next-hop address associated with this route.

The exact next-hop IP address or interface cannot be determined because it is not provided in the given routing table.

Step 4: Determine the Default Route

The default IPv4 route is:

0.0.0.0/0

It is used when no specific route matches the destination IP address and a default route is configured.

If neither a matching route nor a usable default route exists, the router generally drops the packet.

Final Answer

Task	**Answer**
Destination IP Address	192.168.2.45
Destination Subnet	192.168.2.0/24
Matching Route	192.168.2.0/24
Next-Hop Interface	Associated interface for the matching route; exact value not provided
Default Route	0.0.0.0/0

Conclusion: The destination address 192.168.2.45 belongs to 192.168.2.0/24. The router uses the matching route to forward the packet. If no specific route matches, it can use the configured default route 0.0.0.0/0.

QUESTION 5: IP Address Allocation for Administration and Finance LANs

Given Data

Administration LAN: 192.168.10.0/26

Finance LAN: 192.168.10.64/26

Administration Gateway: 192.168.10.1

Finance Gateway: 192.168.10.65

Subnet Mask: 255.255.255.192

Step 1: Calculate the Subnet Size

Prefix length = /26

Host bits = 32 − 26 = 6

Total IP addresses = 2^6 = 64

Usable host addresses = 64 − 2 = **62**

Block size = 256 − 192 = 64

Each LAN supports 62 usable host addresses.

Step 2: Administration LAN Details

Network Address: 192.168.10.0/26

- Network Address: 192.168.10.0
- Valid Host IP Range: 192.168.10.1 – 192.168.10.62
- Broadcast Address: 192.168.10.63
- Subnet Mask: 255.255.255.192
- Default Gateway: 192.168.10.1
- Usable Hosts: 62

Step 3: Finance LAN Details

Network Address: 192.168.10.64/26

- Network Address: 192.168.10.64
- Valid Host IP Range: 192.168.10.65 – 192.168.10.126
- Broadcast Address: 192.168.10.127
- Subnet Mask: 255.255.255.192
- Default Gateway: 192.168.10.65
- Usable Hosts: 62

Step 4: Assign IP Addresses to Three Computers in Each LAN

Administration LAN

Device	IP Address	Subnet Mask	Default Gateway
Computer 1	192.168.10.2	255.255.255.192	192.168.10.1
Computer 2	192.168.10.3	255.255.255.192	192.168.10.1
Computer 3	192.168.10.4	255.255.255.192	192.168.10.1

Finance LAN

Device	IP Address	Subnet Mask	Default Gateway
Computer 1	192.168.10.66	255.255.255.192	192.168.10.65

Computer 2 192.168.10.67 255.255.255.192 192.168.10.65

Computer 3 192.168.10.68 255.255.255.192 192.168.10.65

All six computer addresses are valid host addresses and do not conflict with the configured gateway addresses.

Step 5: Identify the Default Gateways

Administration LAN Default Gateway: **192.168.10.1**

Finance LAN Default Gateway: **192.168.10.65**

Hosts use their respective router interfaces as the default gateway for communication with other networks.

Final Answer

Details	**Administration LAN**	**Finance LAN**
Network Address	192.168.10.0	192.168.10.64
Subnet Mask	255.255.255.192	255.255.255.192
Valid Host Range	192.168.10.1–62	192.168.10.65–126
Broadcast Address	192.168.10.63	192.168.10.127
Default Gateway	192.168.10.1	192.168.10.65
Usable Hosts	62	62

Conclusion: The Administration and Finance LANs use separate /26 subnets. Each LAN supports 62 usable host addresses. The Layer-3 router enables inter-subnet communication when both interfaces and routing are correctly configured.

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand